AKSHIT SRIVASTAVA

+91-8104394240 | as56intn@gmail.com | LinkedIn | GitHub | LeetCode

EDUCATION

Birla Institute of Technology, Mesra
Bachelor of Technology in Computer Science & Engineering — Expected May 2027

EXPERIENCE

Research Intern – IIIT Nagpur — May – Aug 2026
Hyperspectral Imaging — Python, PyTorch, Scikit-learn

- Built reusable preprocessing and learning pipelines for PIKA-NIR hyperspectral imagery spanning 168 spectral bands and six soil classes.
- Developed physics-guided models using spectral derivatives, SVMs, and deep neural networks with leakage-free, cube-level evaluation.
- Co-authored research proposing PG-SD-SVM (99.12% accuracy, 99.04% Macro-F1) and a second paper on explainable Transformer-based spectral prediction, both under review (ICVGIP 2026, CVIP 2026).

Machine Learning Intern – Infosys Springboard — Feb – Apr 2026
TransferIQ — Python, FastAPI, XGBoost, LSTM

- Built and shipped an end-to-end prediction pipeline from scratch, combining LSTM-derived features with XGBoost for a football transfer-value model.
- Designed preprocessing, feature engineering, model training, and inference workflows as modular, reusable components.
- Exposed model inference through a FastAPI service and delivered a deployable production-ready application.

AI Intern – Edunet Foundation — Jun – Jul 2025
Carbon Emission Prediction — Python, Pandas, Scikit-learn

- Built an end-to-end ML pipeline for CO2 emission prediction using environmental and climate-related data.
- Applied preprocessing, feature scaling, cross-validation, and ensemble regression for predictive modeling.

PROJECTS

AI Career Mentor — GitHub | Live
Tech: Python, React, Groq, Cloudflare Workers, D1.
Built an AI-powered career platform from scratch that evaluates resumes, grades role fit, identifies skill gaps, and generates personalized learning roadmaps. Integrated Groq-hosted LLM inference with a native Python Cloudflare Worker to generate structured, role-specific recommendations, and implemented persistent assessment history and roadmap tracking using Cloudflare D1.

Checkout + Inventory Platform — GitHub
Tech: Python, FastAPI, PostgreSQL, SQLAlchemy, React, JWT.
Built a full-stack e-commerce platform with a FastAPI/PostgreSQL backend and React frontend, handling authentication, catalog, cart, and order management. Designed atomic checkout using PostgreSQL row-level locking to prevent overselling under concurrent orders, and modeled order lifecycle transitions across pending, paid, shipped, and delivered states. Implemented JWT-based auth with role-based access control, real-time catalog updates via Server-Sent Events, and REST APIs documented with Swagger/OpenAPI.

SKILLS

Languages: Python, Java, C, SQL, JavaScript
Backend & APIs: FastAPI, Flask, REST APIs, Cloudflare Workers
Machine Learning: PyTorch, Scikit-learn, XGBoost, LightGBM, SVM, LSTM, CNNs, PCA, SHAP
AI & LLM Integration: Groq, LLM Integration, NLP, VADER, TextBlob
Data & Scientific: NumPy, Pandas, Matplotlib, SciPy, Scikit-image
Databases: PostgreSQL, MySQL, Supabase, Cloudflare D1
Tools & Deployment: Git, Docker, Linux, MLflow, Jupyter, Vercel, Render

CERTIFICATIONS & RECOMMENDATIONS

- Research Internship Certificate – Indian Institute of Information Technology, Nagpur
- Letter of Recommendation – Indian Institute of Information Technology, Nagpur
- Machine Learning Internship Certificate – Infosys Springboard 6.0